A finite-ensemble obstruction to Holevo-rate visible compression

Abstract

We construct a finite ensemble of rank-eight states on $\mathbb{C}^9$ whose visible quantum compression rate is strictly larger than its Holevo information, even with arbitrary finite shared classical randomness at each blocklength. The model excludes initially shared entanglement and averages over the shared seed before the trace-norm error. A finite-block inequality gives a strong converse and a strict rate gap above $\log_2(9/8)$.

All logarithms below are base two. The shared seed is independent of the source and is averaged before the trace norm, as specified below.

1 Statement

The coding model permits shared classical randomness and a quantum message, but no initially shared entanglement or additional communication. The shared seed can have arbitrary finite size at each blocklength.

A length-n visible code has a message space M_n of dimension d_n, a finite shared seed s of source-independent probability $q_n(s)$, encoded density matrices $\tau_{x^n,s}$ on M_n, and CPTP decoding maps $\mathcal{D}_{n,s}$. Its reproduced state and error are

$$\widehat{W}_{x^n} = \sum_s q_n(s)\mathcal{D}_{n,s}(\tau_{x^n,s}),$$
$$\varepsilon_n = \sum_{x^n} p^n(x^n)\frac{1}{2}\|\widehat{W}_{x^n} - W_{x^n}\|_1.$$

The rate $R_{\mathrm{vis},q,\mathrm{cr}}$ is the infimum of $\limsup_n n^{-1}\log_2 d_n$ over sequences of such codes with $\varepsilon_n \to 0$. The Holevo information is $\chi(p,W) = S(\sum_x p_x W_x) - \sum_x p_x S(W_x)$, with $S(\rho) = -\operatorname{Tr}(\rho\log_2\rho)$.

There is an explicit finite ensemble of rank-eight mixed states on $\mathbb{C}^9$, with full-support probabilities, for which

$$\chi(p,W) = \log_2(9/8) \quad \text{and} \quad R_{\mathrm{vis},q,\mathrm{cr}}(p,W) \geq \frac{2}{3}\log_2(45/37) = \log_2(9/8) + \frac{1}{3}\log_2(12800/12321). \tag{A}$$

The last summand is strictly positive. Thus the identity $R_{\mathrm{vis},q,\mathrm{cr}}(p,W) = \chi(p,W)$ does not hold for every finite ensemble in this model. This contradicts the communication-rate assertion of Winter's Conjecture 23 [Win02, Section VII, p. 11], even after removing its bound on the amount of shared randomness. Equations (15)–(16) of [Win02] define the unassisted quantum visible-coding model. Section III introduces shared randomness for classical sources; Conjecture 23 proposes the quantum extension. Here the reproduced state is averaged over that randomness before the error norm.

Hayashi [Hay06, Section 10.8, Lemma 10.5] gives the Holevo lower bound in the shared-randomness model. Equality in his Theorem 10.6 assumes commuting signal states. The noncommuting ensemble constructed below lies outside that hypothesis.

In fact, every n-block code for this ensemble, with $d = \dim M_n$ and the seed-averaged block error ε_n, obeys

$$(1-\varepsilon_n)^2 \leq d^{3/2}(37/45)^n. \tag{B}$$

Thus the lower bound is a strong converse.

2 An explicit finite ensemble

Let $\zeta = e^{2\pi i/3}$, and fix the standard basis $e_1, \dots, e_9$. There are two families of labels:

- The nine basis vectors e_j, each with probability $1/90$.
- For every $r = (r_2, \dots, r_9) \in \{0,1,2\}^8$, the vector

$$v_r = \frac{1}{3}\left(e_1 + \sum_{j=2}^{9} \zeta^{r_j} e_j\right),$$

  each with probability $1/7290$.

There are $9 + 3^8 = 6570$ labels, all with positive probability. Their probabilities sum to $9/90 + 3^8/7290 = 1/10 + 9/10 = 1$. For the vector u_x associated with a label, put

$$Q_x = |u_x\rangle\langle u_x|, \qquad P_x = I_9 - Q_x, \qquad W_x = P_x/8. \tag{C}$$

Every W_x is a density operator of rank eight. This one-letter dimension, label set and probability distribution are fixed, independent of n.

Write F for the swap on $\mathbb{C}^9 \otimes \mathbb{C}^9$, and $Z = \sum_{j=1}^{9} |jj\rangle\langle jj|$. The pure projectors have moments

$$\mathbb{E}_x Q_x = I_9/9, \qquad \mathbb{E}_x (Q_x \otimes Q_x) = \frac{I + F}{90}. \tag{D}$$

Here is a direct calculation of the second identity. In the phase family, set $r_1 = 0$. Its matrix entry with row indices (i,j) and column indices (k,l) is

$$\frac{1}{81}\mathbb{E}_r \zeta^{r_i + r_j - r_k - r_l}.$$

For every coordinate $h \geq 2$, the exponent of r_h is an integer in $[-2,2]$. Averaging over a cube root gives zero unless that exponent is zero. If all these exponents vanish, the exponent for coordinate 1 vanishes as well, since the total exponent over coordinates is zero. Therefore the entry is $1/81$ precisely when the index multisets $\{i,j\}$ and $\{k,l\}$ agree, and is zero otherwise. Equivalently,

$$\mathbb{E}_{\text{phase}}(Q \otimes Q) = \frac{I + F - Z}{81}.$$

The basis-family moment is $Z/9$. With respective total weights $9/10$ and $1/10$, these give

$$\frac{9}{10}\frac{I + F - Z}{81} + \frac{1}{10}\frac{Z}{9} = \frac{I + F}{90}.$$

Taking a partial trace gives the first identity in (D), because $\mathrm{Tr}_2\, I = 9I$ and $\mathrm{Tr}_2\, F = I$. No enumeration of the phase vectors is needed.

It follows from (D) and (C) that

$$\mathbb{E}_x W_x = I_9/9, \qquad \chi(p, W) = \log_2 9 - \log_2 8 = \log_2(9/8). \tag{E}$$

Expanding $(I - Q_x) \otimes (I - Q_x)$ gives

$$\mathbb{E}_x (P_x \otimes P_x) = aI + bF, \qquad a = \frac{71}{90}, \quad b = \frac{1}{90}. \tag{F}$$

Indeed, the coefficient of I is $1 - 2/9 + 1/90 = 71/90$.

3 A channel-dimension lemma

Let $\Psi : \mathcal{L}(\mathbb{C}^d) \to \mathcal{L}(\mathbb{C}^h)$ be any CPTP map. For a fixed basis of its input, let $E_{ij} = |i\rangle\langle j|$. Its unnormalized Choi operator is

$$J_\Psi = \sum_{i,j=1}^{d} E_{ij} \otimes \Psi(E_{ij}).$$

Complete positivity and trace preservation give

$$J_\Psi \geq 0, \qquad \mathrm{Tr}_{\mathrm{out}}\, J_\Psi = I_d, \qquad \mathrm{Tr}\, J_\Psi = d. \tag{G}$$

We claim

$$\mathrm{Tr}\, J_\Psi^2 \leq d \min\{d, h\}. \tag{H}$$

Positivity and trace d give $\|J_\Psi\|_\infty \leq d$, so $\mathrm{Tr}\, J_\Psi^2 \leq d^2$.

For the other bound, write $J_\Psi = (J_{bb'})_{b,b'=1}^{h}$ in an output basis. The positive diagonal blocks sum to I_d, so $0 \leq J_{bb} \leq I_d$. For a vector $v = \sum_b v_b \otimes |b\rangle$, positivity gives

$$|\langle v_b, J_{bb'} v_{b'}\rangle| \leq \sqrt{\langle v_b, J_{bb} v_b\rangle}\sqrt{\langle v_{b'}, J_{b'b'} v_{b'}\rangle}.$$

This follows by applying Cauchy–Schwarz to $J_\Psi^{1/2}(v_b \otimes |b\rangle)$ and $J_\Psi^{1/2}(v_{b'} \otimes |b'\rangle)$. Consequently

$$\begin{aligned}\langle v, J_\Psi v\rangle &\leq \left(\sum_b \sqrt{\langle v_b, J_{bb} v_b\rangle}\right)^2 \\ &\leq h \sum_b \langle v_b, J_{bb} v_b\rangle \leq h \sum_b \|v_b\|^2 = h\|v\|^2.\end{aligned}$$

Thus $J_\Psi \leq hI$ and $\mathrm{Tr}\, J_\Psi^2 \leq h\, \mathrm{Tr}\, J_\Psi = hd$. Combining the two bounds proves (H).

This lemma applies to every channel, with arbitrary ancillary environments already included in the map.

4 A tensor-product support-score bound

Fix any CPTP decoder $\Phi : \mathcal{L}(\mathbb{C}^d) \to \mathcal{L}((\mathbb{C}^9)^{\otimes n})$. For x^n, put $P_{x^n} = \bigotimes_{t=1}^n P_{x_t}$ and

$$A_{x^n} = \Phi^*(P_{x^n}), \qquad L(\Phi) = \mathbb{E}_{x^n} \lambda_{\max}(A_{x^n}).$$

Since Φ^* is positive and unital, $0 \leq A_{x^n} \leq I_d$. For any density operator τ_{x^n},

$$\mathbb{E}_{x^n}\, \mathrm{Tr}[P_{x^n} \Phi(\tau_{x^n})] \leq L(\Phi). \tag{I}$$

Cauchy–Schwarz and the eigenvalue inequality give

$$L(\Phi)^2 \leq \mathbb{E}_{x^n} \lambda_{\max}(A_{x^n})^2 \leq \mathbb{E}_{x^n}\, \mathrm{Tr}\, A_{x^n}^2. \tag{J}$$

We compute the last expression explicitly. The adjoint identity implies

$$[A_{x^n}]_{ij} = \mathrm{Tr}[P_{x^n} \Phi(E_{ji})],$$

and therefore

$$\operatorname{Tr} A_{x^n}^2 = \sum_{i,j} \operatorname{Tr}[P_{x^n}\Phi(E_{ij})] \operatorname{Tr}[P_{x^n}\Phi(E_{ji})]. \tag{K}$$

Let $F_d = \sum_{i,j} E_{ij} \otimes E_{ji}$ be the input swap. Equation (K) is equivalent to

$$\operatorname{Tr} A_{x^n}^2 = \operatorname{Tr}\left[(\Phi \otimes \Phi)(F_d)(P_{x^n} \otimes P_{x^n})\right]. \tag{L}$$

For each $T \subseteq [n]$, let F_T swap the two output copies at positions in T and act as identity elsewhere. The source is i.i.d.; hence (F) implies

$$\mathbb{E}_{x^n}(P_{x^n} \otimes P_{x^n}) = \prod_{t=1}^{n}(aI + bF_{\{t\}}) = \sum_{T\subseteq[n]} a^{n-|T|}b^{|T|}F_T. \tag{M}$$

This is the moment of product signals from the fixed finite ensemble. It does not assert that the n-letter supports form a Haar ensemble on the whole output space.

Let $\Phi_T = \operatorname{Tr}_{[n]\setminus T} \circ \Phi$. It is CPTP from a d-dimensional input to a $9^{|T|}$-dimensional output. For $T = \varnothing$, take its output to be the one-dimensional scalar system. The swap identity $\operatorname{Tr}[(X \otimes Y)F] = \operatorname{Tr}(XY)$ gives

$$\begin{aligned} \operatorname{Tr}[(\Phi \otimes \Phi)(F_d)F_T] &= \sum_{i,j} \operatorname{Tr}[\Phi_T(E_{ij})\Phi_T(E_{ji})] \\ &= \operatorname{Tr} J_{\Phi_T}^2. \end{aligned} \tag{N}$$

The second equality follows by multiplying the matrix units in $J_{\Phi_T} = \sum_{i,j} E_{ij} \otimes \Phi_T(E_{ij})$: $\operatorname{Tr}(E_{ij}E_{kl}) = \delta_{jk}\delta_{il}$. In particular, no partial transpose is present. When $T = \varnothing$, $J_{\Phi_T} = I_d$ and this term is exactly d.

Combining (L)–(N) and the channel lemma,

$$\begin{aligned} \mathbb{E}_{x^n} \operatorname{Tr} A_{x^n}^2 &= \sum_{T\subseteq[n]} a^{n-|T|}b^{|T|} \operatorname{Tr} J_{\Phi_T}^2 \\ &\le d \sum_{T\subseteq[n]} a^{n-|T|}b^{|T|} \min\{d, 9^{|T|}\} \\ &\le d^{3/2} \sum_{T\subseteq[n]} a^{n-|T|}b^{|T|}3^{|T|} \\ &= d^{3/2}(a+3b)^n = d^{3/2}(37/45)^n. \end{aligned} \tag{O}$$

Here $\min\{d, h\} \le \sqrt{dh}$ for positive d, h. All expansion coefficients $a^{n-|T|}b^{|T|}$ are nonnegative. The operator $(\Phi \otimes \Phi)(F_d)$ need not be positive; no order comparison involving it was used. Instead each scalar term in (N) is exactly a nonnegative Choi purity.

Equations (J) and (O) establish

$$L(\Phi) \le d^{3/4}(37/45)^{n/2}. \tag{P}$$

5 Arbitrary free shared randomness and the exact error criterion

For an arbitrary code with source-independent seed probabilities $q_n(s)$, encoded states $\tau_{x^n,s}$ and CPTP decoders $D_{n,s}$, put

$$\begin{aligned} V_n &= \sum_{x^n} p^n(x^n) \operatorname{Tr}[P_{x^n}\widehat{W}_{x^n}] \\ &= \sum_s q_n(s) \sum_{x^n} p^n(x^n) \operatorname{Tr}[P_{x^n}\mathcal{D}_{n,s}(\tau_{x^n,s})]. \end{aligned} \tag{Q}$$

For each seed, (I) and (P) apply with the same $d = \dim M_n$. Because the seed is independent of the labels, the conditional source distribution is p^n. Thus, without any restriction on the finite seed's size or probabilities,

$$V_n \le \sum_s q_n(s) L(\mathcal{D}_{n,s}) \le d^{3/4}(37/45)^{n/2}. \tag{R}$$

The target $W_{x^n} = P_{x^n}/8^n$ has $\mathrm{Tr}[P_{x^n} W_{x^n}] = 1$. For two trace-one states ρ, σ and any effect $0 \le E \le I$,

$$|\,\mathrm{Tr}[E(\rho - \sigma)]| \le \tfrac{1}{2}\|\rho - \sigma\|_1.$$

For example, this follows by writing the traceless Hermitian difference as its positive part minus its negative part, each of trace $\frac{1}{2}\|\rho - \sigma\|_1$. Apply it to the effect P_{x^n} and the seed-averaged state $\widehat{W}_{x^n}$. Averaging over labels gives

$$V_n \ge 1 - \varepsilon_n. \tag{S}$$

The error is computed after averaging the seed; no error condition on individual seeds is used.

Equations (R) and (S), followed by squaring nonnegative quantities, prove (B).

6 The strict asymptotic gap

For a vanishing-error code, eventually $\varepsilon_n < 1$. Taking logarithms in (B) gives

$$\frac{1}{n}\log_2 d \ge \frac{2}{3}\log_2(45/37) + \frac{4}{3n}\log_2(1 - \varepsilon_n).$$

The final term tends to zero, so even the lower limit of every valid code's rate is at least the first term. Taking the infimum of upper-limit rates proves the lower bound in (A).

Finally,

$$\begin{aligned}\frac{2}{3}\log_2(45/37) - \log_2(9/8) &= \frac{1}{3}\log_2\left[\frac{(45/37)^2}{(9/8)^3}\right] \\ &= \frac{1}{3}\log_2(12800/12321) > 0,\end{aligned}$$

since $12800 > 12321$. This proves the strict gap for a fixed finite mixed-state ensemble, allowing every independent finite shared seed and every CPTP decoder in the stated model.

7 References

- **[Win02]** A. Winter, *Compression of sources of probability distributions and density operators* (2002), arXiv:quant-ph/0208131, Sections III and VII, Eqs. (15)–(16) and Conjecture 23.
- **[Hay06]** M. Hayashi, *Quantum Information: An Introduction*, Springer (2006), Section 10.8, pp. 339–341, Lemma 10.5 and Theorem 10.6, DOI.